

DSCI 1411 – Final Project

Data Exploration and Analysis

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1. Introduction

Nowadays, music streaming platforms such as Spotify have become the way people discover and listen to music. Now that there are millions of songs online, understanding what musical characteristics are associated with higher popularity is valuable for artists, producers, and the music industry. Spotify provides detailed audio features for each track, allowing for a data driven exploration of patterns related to song popularity.

About the Data Set:

The dataset we chose provides patterns that show what global music has been trending on Spotify from 2015 to 2025. It studies how genres, popularity, and audio features have changed over a decade. The dataset contains 85000 tracks including both qualitative and quantitative variables, such as genre, country, popularity score, danceability, energy, tempo, duration, and explicit content. It was also designed to help with some machine learning tasks like predicting popularity, classify genres, cluster tracks, or perform regression analysis.

Problem statement :

This project aims to explore what musical characteristics are associated with higher song popularity on Spotify and how popularity varies across different genres.

Objectives :

- Summarize the main characteristics of songs on Spotify using descriptive statistics
- Examine the distribution of song popularity and identify potential outliers
- Explore relationships between popularity and musical features such as danceability, energy, and tempo
- Compare popularity across different music genres

2. Methodology

The dataset was explored by some analysis using some descriptive statistical techniques. The data was then imported to R and started selecting the most relevant variables. Both quantitative and qualitative variables were analyzed in graphical, tabular and numerical methods.

Tabular techniques were used to summarize categorical variables. Frequency tables were created to show the distribution of songs across genres, and cross tabulations were used to examine the relationship between genre and whether a song is explicit.

Graphical techniques were used to visually explore patterns and relationships in the data. Histograms were used to examine the distribution of popularity scores. Bar charts were used to compare the number of songs across genres. Boxplots were used to compare popularity across genres and to identify outliers. Scatter plots were used to explore relationships between popularity and continuous musical features such as danceability.

Numerical methods were applied to summarize quantitative variables. Measures of central tendency (mean and median) and measures of dispersion (standard deviation, variance, minimum, and maximum) were calculated for popularity, danceability, energy, and tempo.

These methods were chosen because they provide a clear and effective way to describe the data, identify trends and outliers, and relate findings back to the project's problem statement.

3. Analysis

3.1 Descriptive Statistics

Summary statistics were calculated for the main quantitative variables in the dataset, including popularity, danceability, energy, and tempo. These statistics provide an overview of the typical values and variability within the data.

Table 1: Frequency Distribution of Songs by Genre

```
> table(spotify$genre)
```

Classical	Country	EDM	Folk
7158	7030	6894	7080
Hip-Hop	Indie	Jazz	Metal
7160	7007	7177	7200
Pop	R&B	Reggaeton	Rock
7096	7084	7001	7113

Table 2: Summary Statistics for Song Popularity

Numerical methods to summarize the data, such as calculating measures of central tendency (mean, median, mode) and dispersion (variance, standard deviation).

```
> mean(spotify$popularity)
[1] 48.16289
> median(spotify$popularity)
[1] 47
> sd(spotify$popularity)
[1] 14.82975
> var(spotify$popularity)
[1] 219.9215
> min(spotify$popularity)
[1] 0
> max(spotify$popularity)
[1] 100
```

The mean popularity score gives an indication of how popular songs in the dataset are on average, while the standard deviation shows how much popularity

varies across songs. Similar interpretations apply to danceability, energy, and tempo, which describe the musical characteristics of the tracks.

Summary tables:

```
> summary(spotify)
  track_id      track_name      artist_name      album_name      release_date
Length:85000   Length:85000   Length:85000   Length:85000   Length:85000
Class :character Class :character Class :character Class :character Class :character
Mode  :character Mode  :character Mode  :character Mode  :character Mode  :character

  genre      duration_ms      popularity      danceability      energy
Length:85000 Min.   : 90004 Min.   :  0.00 Min.   :0.0500 Min.   :0.0200
Class :character 1st Qu.:171872 1st Qu.: 38.00 1st Qu.:0.2900 1st Qu.:0.2600
Mode  :character Median :254920 Median : 47.00 Median :0.5200 Median :0.5000
                Mean  :254914 Mean  : 48.16 Mean  :0.5207 Mean  :0.5054
                3rd Qu.:337203 3rd Qu.: 57.00 3rd Qu.:0.7600 3rd Qu.:0.7500
                Max.   :420000 Max.   :100.00 Max.   :0.9900 Max.   :0.9900

  key      loudness      mode      instrumentalness      tempo
Min.   : 0.000 Min.   : -55.00 Min.   :0.0000 Min.   :0.0000 Min.   : 60.00
1st Qu.: 2.750 1st Qu.: -41.47 1st Qu.:0.0000 1st Qu.:0.1980 1st Qu.: 94.84
Median : 6.000 Median : -27.97 Median :1.0000 Median :0.3990 Median :129.99
Mean   : 5.516 Mean   : -28.00 Mean   :0.5001 Mean   :0.3998 Mean   :129.95
3rd Qu.: 9.000 3rd Qu.: -14.50 3rd Qu.:1.0000 3rd Qu.:0.6000 3rd Qu.:165.03
Max.   :11.000 Max.   : -1.00 Max.   :1.0000 Max.   :0.8000 Max.   :200.00

  stream_count      country      explicit      label
Min.   :   1000 Length:85000 Min.   :0.0000 Length:85000
1st Qu.:   1000 Class :character 1st Qu.:0.0000 Class :character
Median :   2000 Mode  :character Median :0.0000 Mode  :character
Mean   :  214355              Mean  :0.2013
3rd Qu.:   9000              3rd Qu.:0.0000
Max.   :20000000              Max.   :1.0000
```

```
> summary(spotify[, c("popularity", "danceability", "energy", "tempo", "duration_ms")])
  popularity      danceability      energy      tempo      duration_ms
Min.   :  0.00 Min.   :0.0500 Min.   :0.0200 Min.   : 60.00 Min.   : 90004
1st Qu.: 38.00 1st Qu.:0.2900 1st Qu.:0.2600 1st Qu.: 94.84 1st Qu.:171872
Median : 47.00 Median :0.5200 Median :0.5000 Median :129.99 Median :254920
Mean   : 48.16 Mean   :0.5207 Mean   :0.5054 Mean   :129.95 Mean   :254914
3rd Qu.: 57.00 3rd Qu.:0.7600 3rd Qu.:0.7500 3rd Qu.:165.03 3rd Qu.:337203
Max.   :100.00 Max.   :0.9900 Max.   :0.9900 Max.   :200.00 Max.   :420000
```

Table 3: CrossTabulation of Genre and Explicit Content

```
> table(spotify$genre, spotify$explicit)
```

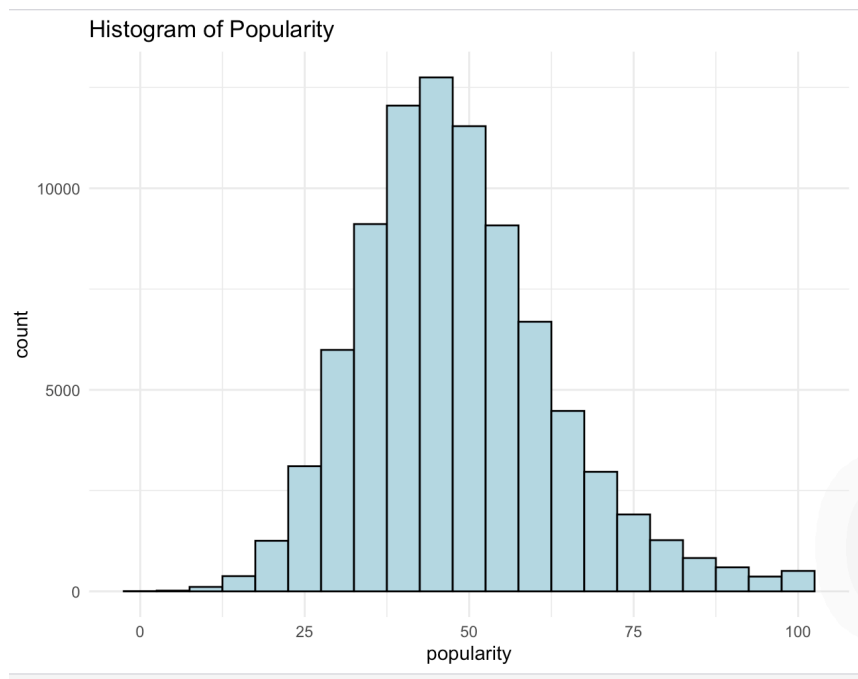
	0	1
Classical	5678	1480
Country	5617	1413
EDM	5561	1333
Folk	5630	1450
Hip-Hop	5686	1474
Indie	5583	1424
Jazz	5805	1372
Metal	5749	1451
Pop	5652	1444
R&B	5667	1417
Reggaeton	5606	1395
Rock	5653	1460

```
##
```

3.2 Distribution of Song Popularity

A histogram was created to visualize the distribution of song popularity. This graph helps identify whether popularity scores are evenly distributed or skewed, and whether there are extreme values.

Figure 1: Histogram showing the distribution of song popularity on Spotify.

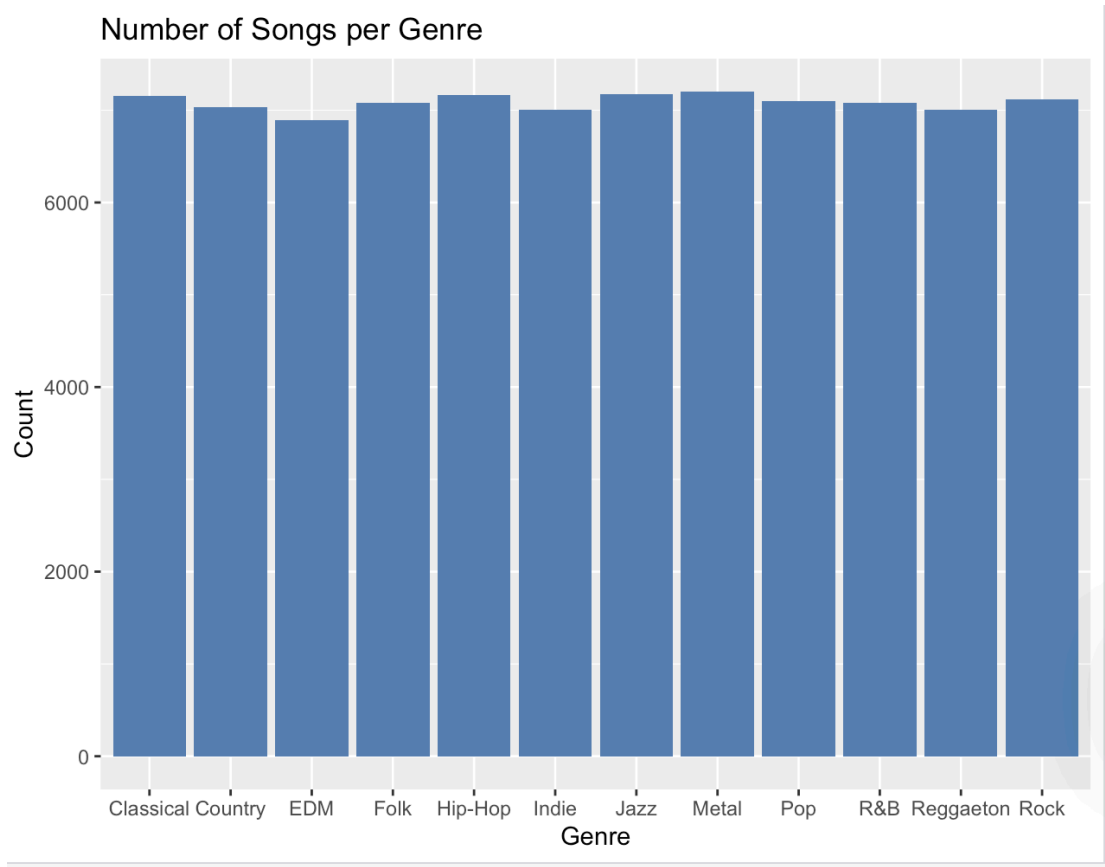


This histogram is skewed to the right, asymmetrical and unimodal. It shows how song popularity is not evenly distributed. Most songs fall within a lower to middle popularity range, while fewer songs achieve very high popularity. This indicates that highly popular songs are relatively rare, suggesting a right skewed distribution.

3.3 Genre Distribution

A bar chart was used to show the number of songs in each genre.

Figure 2: Bar chart displaying the number of songs per genre.

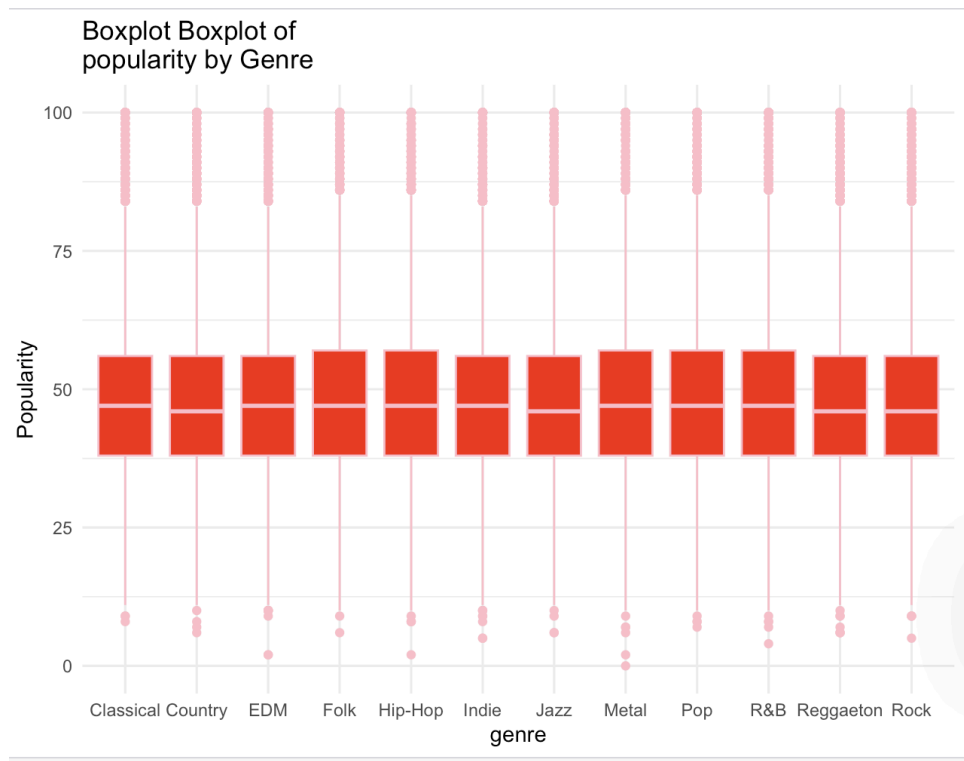


The bar chart shows that some genres are different than others in the dataset. Some have more count. This imbalance is important to consider when comparing popularity across genres, as genres with more songs may show greater variability.

3.4 Popularity by Genre

A boxplot was created to compare song popularity across different genres.

Figure 3: Boxplot comparing song popularity across genres.



The boxplot reveals minimal differences in popularity between genres. The medians seem to be rather similar which means that there is no relation between the genres and their popularity. But, if we do take in consideration the small differences we can conclude that some genres have higher median popularity scores, while others show a wider spread, indicating greater variability.

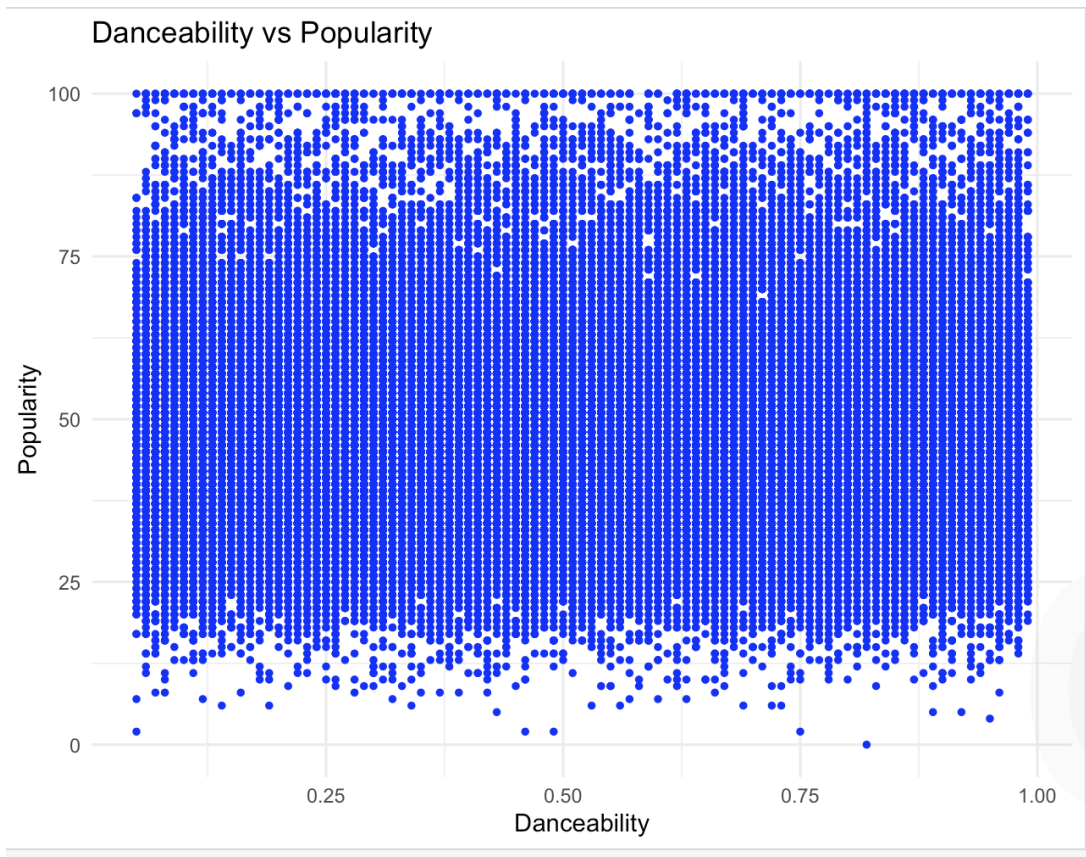
3.5 Outlier Analysis

Outliers in popularity were identified using the interquartile range (IQR) method. They are represented by dots. Using this approach, a small number of songs (approximately 5–10 tracks) were identified as outliers with popularity scores significantly higher or lower than the rest of the dataset. This means that certain songs perform exceptionally well or poorly compared to others within the same genre. Identifying outliers is important because they can influence summary statistics and highlight exceptional cases within the data.

3.6 Relationship Between Danceability and Popularity

A scatter plot was used to examine the relationship between danceability and popularity.

Figure 4: Scatter plot of danceability versus popularity.



The scatter plot suggests a weak positive relationship between danceability and popularity. While higher danceability does not guarantee high popularity, songs with higher danceability tend to appear more frequently among moderately to highly popular tracks.

4. Conclusion

This project explored patterns in Spotify song data using descriptive statistical methods. The analysis showed that song popularity is unevenly distributed, with most tracks achieving moderate popularity and only a few reaching very high popularity levels. Differences in popularity were observed across genres, suggesting that genre plays a role in how songs are received by listeners.

The relationship between danceability and popularity appears to be positive but weak, which indicates that while musical features may contribute to popularity, they are not the only determining factors. Other influences such as marketing, artist reputation, and social trends likely also play a significant role.

Overall, this analysis demonstrates how descriptive statistics and data visualization can be used to gain meaningful insights from real-world datasets. A limitation of this study is that it focuses only on Spotify audio features and does not account for external factors affecting popularity. Future analyses could include additional variables such as release year or artist popularity to gain deeper insights.

5. Reference:

<https://www.kaggle.com/datasets/rohiteng/spotify-music-analytics-dataset-20152025>